

Subset Row Cut (SRC) — Strengthening the LP Relaxation

An SRC is associated with a subset $C \subseteq V$ of customers.

For a set C with $|C| \geq 3$ customers, the SRC reads:

$$\sum_{r \in \Omega} \lfloor \frac{1}{2} \rfloor \sum_{i \in C} a_{ir} \lambda_r \leq \lfloor \frac{|C|}{2} \rfloor$$

where $a_{ir} = 1$ if route r serves customer i , and λ_r is the route variable.

The cut exploits the fact that a vehicle visiting an odd subset C more times than $\lfloor |C|/2 \rfloor$ would need to re-enter C , which is suboptimal for the LP.